Shoulder Recognition Algorithm Documentation

Real Galaxy Shoulder Quantification v0.69

Field	Value
Script	Real Galaxy Shoulder Quantification v0.69.py
Location	C:\Users\gordo\Documents\Github\PythonScripts\Shoulder Recognition Erwin\Real Galaxy Shoulder Quantification v0.69.py
Document date	2026-07-03
Documentation scope	High-level overview, data sources, processing workflow, outputs, workbook merge/format utilities, parameters, and detailed code behavior.

High-Level Overview

This script runs a shoulder-quantification workflow on observed galaxy bar major-axis profiles. Its main purpose is to identify paired shoulder structures in bar profiles from the Erwin, Debattista, and Anderson 2023 classification sample, using S4G 3.6 micron FITS images and a locally generated geometry manifest to construct the required profiles when profile files are not already available.

For each classified galaxy with the required catalog, geometry, and image data, the code extracts a deprojected bar-major-axis intensity profile, normalises radius by the deprojected bar radius, smooths the analysed profile with a low-pass Butterworth filter, computes first and second derivatives and radius-of-curvature structure, then searches for left and right shoulder candidates inside the bar.

- Input data are external to this folder and are selected with the --pc Laptop/Desktop switch for the two Dropbox layouts.
- The analysis window is restricted to $|x/R_{bar}| < 1.8$, while accepted shoulders must lie inside $|x/R_{bar}| < 1$.
- By default, a galaxy is counted as having shoulders only when both left and right shoulders are accepted.

- The script writes constructed profile caches, missing-data reports, measurement tables, a per-galaxy classification CSV, and a run summary.
- Diagnostic PNGs are not regenerated unless --write-plots is supplied.

Updates Added 2026-06-25

The shoulder-recognition workflow was extended to support the two working-machine Dropbox layouts, preserve existing plots by default, export explicit SRA classifications by galaxy, and merge those results into the PE/VPD workbook.

- Real Galaxy Shoulder Quantification v0.69.py now accepts --pc Laptop or --pc Desktop. Laptop maps to C:\Users\gordo\Dropbox\Public Documents\UCLAN\MSc Research; Desktop maps to D:\Dropbox\Public Documents\UCLAN\MSc Research.
- Diagnostic plot PNGs are skipped by default. Use --write-plots only when plots/<galaxy>.png should be rebuilt.
- The analysis now writes shoulder_classifications.csv with one row per galaxy and explicit labels: Shoulders, No Shoulders, Too Noisy, or Missing Data.
- Merge Shoulder Classifications Into Workbook.py updates PE_VPD_galaxy_classifications_with_definitions.xlsx from shoulder_classifications.csv and creates a timestamped workbook backup before saving.
- The merge utility also adds PE/VPD profile-class columns from the same classifications_pe.txt and classifications_vd_revised.txt files used by barprofiles_figures_for_paper.py.
- Format PE VPD Classification Workbook.py reorders the workbook so PE/VPD classifications are on the left and SRA output columns are grouped on the right. It applies bold white text on a dark-blue header background, freezes the top row, enables filters, auto-sizes columns, and creates a timestamped backup.
- The formatted workbook now adds an isophote profile PDF hyperlink column plus GB visual class and GB visual notes columns. The visual class column uses a dropdown with Peak+Sh, Exp, Flat-top (FT), Two-slope (2S), and Unclear.
- Build Bar Profile Visual Gallery.py creates a local HTML gallery for browsing the same isophote profile PDFs with PE/VPD/SRA context, visual-class dropdowns, notes, search, and CSV export from the browser.
- The gallery has a Reveal Classifiers toggle. When it is off, PE, VPD, and SRA labels are all hidden for every galaxy. When it is on, all three classifier labels are shown.
- The gallery can also be run with --serve, which starts a local server and writes GB visual class and GB visual notes changes directly back to the workbook. Laptop and Desktop launchers use http://127.0.0.1:8899/.

Command	Purpose
python "Shoulder Recognition Erwin\Real Galaxy Shoulder Quantification v0.69.py" --pc Laptop	Run the full shoulder analysis on the Laptop Dropbox path without regenerating plot PNGs.
python "Shoulder Recognition Erwin\Real Galaxy Shoulder Quantification v0.69.py" --pc Desktop	Run the same analysis on the Desktop Dropbox path.
python "Shoulder Recognition Erwin\Real Galaxy Shoulder Quantification v0.69.py" --pc Laptop --write-plots	Run the analysis and regenerate diagnostic plot PNGs.
python "Shoulder Recognition Erwin\Merge Shoulder Classifications Into Workbook.py" --pc Laptop	Merge shoulder_classifications.csv plus PE/VPD profile labels into the PE/VPD workbook.
python "Shoulder Recognition Erwin\Format PE VPD Classification Workbook.py" --pc Laptop	Reorder and style the PE/VPD workbook after merging.
python "Shoulder Recognition Erwin\Build Bar Profile Visual Gallery.py" --pc Laptop	Build the local HTML gallery for visual review of isophote profile PDFs.
python "Shoulder Recognition Erwin\Build Bar Profile Visual Gallery.py" --pc Laptop --serve --port 8899	Run the workbook-writing local gallery server at http://127.0.0.1:8899/.
python "Shoulder Recognition Erwin\Build Bar Profile Visual Gallery.py" --pc Desktop --serve --port 8899	Run the same workbook-writing gallery server using the Desktop Dropbox path.

Data Sources and Dependencies

The script combines local repository helper modules with external research data. The repository supplies geometry and plotting utilities; the measured galaxy data and FITS images live in a hard-coded Dropbox research folder.

Source	Purpose
Erwin catalog data	Reads s4gbars_table.dat for bar and galaxy properties, scrambled_map.txt to map classification IDs to galaxy names, and classifications_pe.txt plus classifications_vd_revised.txt to define the classified sample.
S4G geometry manifest	Reads Erwin_s4g_image_downloader/geometry_output/s4g_image_geometry_manifest.csv for image centres, disk PA, inclination, bar PA, bar semi-major axis, pixel scale, CRPIX fallback values, and image path metadata.
S4G FITS images	Loads 3.6 micron images from each manifest image_path or from <research_folder>\Erwin\s4g_images_36um\<galaxy>.phot.1.fits.
Helper modules	Imports angle_utils from Erwin_barprofiles_paper_GB_working_copy and plot_s4g_isophote_axes from Erwin_s4g_image_downloader for PA rectification, deprojection, and profile extraction utilities.
Scientific Python stack	Uses numpy, pandas, scipy.signal, astropy.io.fits, scikit-image profile_line, matplotlib, csv, math, os, sys, and warnings.

Important hard-coded paths

Name	Value / meaning
--pc Laptop	C:\Users\gordo\Dropbox\Public Documents\UCLAN\MSc Research
--pc Desktop	D:\Dropbox\Public Documents\UCLAN\MSc Research
erwin_repo	<research_folder>\Erwin\perwin-barprofiles_paper-a7cd6f5
erwin_data_folder	<erwin_repo>\data

Name	Value / meaning
manifest_file	<repo>\Erwin_s4g_image_downloader\geometry_output\s4g_image_geometry_manifest.csv
image_folder	<research_folder>\Erwin\s4g_images_36um
output_folder	<research_folder>\Shoulder_Recognition_Erwin
plots_folder	<output_folder>\plots
profiles_folder	<output_folder>\profiles
workbook	<output_folder>\PE_VPD_galaxy_classifications_with_definitions.xlsx

Processing Workflow

1. Initialise paths, create output, plots, and profiles folders, and add local helper-module directories to sys.path.
2. Read the Erwin S4G bar table, the scrambled-name map, the two classification files, and the S4G image geometry manifest.
3. Build a sorted classified-galaxy list by combining Peter Erwin and Victor Debattista revised classifications, excluding unknown classifications marked with '?'.
4. For each classified galaxy, require a catalog row and manifest row, then construct a bar-major-axis profile from the FITS image.
5. Write a missing-data text report and CSV for galaxies that cannot be analysed.
6. Loop over all galaxies with usable profiles, interpolate NaNs, normalise the analysed profile, smooth it, calculate derivatives and radius of curvature, and search for left and right shoulder candidates.
7. Reject shoulders that are too close to the centre, outside the bar, too thin, too steep, too noisy, one-sided when paired shoulders are required, or overlapping.
8. Write final shoulder measurements in NPY and CSV formats, write shoulder_classifications.csv, and write a plain-text run summary.
9. Optionally save one diagnostic PNG per analysed galaxy when --write-plots is supplied.

Profile construction

construct_profile_from_manifest uses the manifest geometry to locate the galaxy centre and bar orientation. It loads the FITS image, falls back to CRPIX coordinates if the manifest centre is outside the image, chooses a profile radius based on the bar semi-major axis and pixel scale, and extracts a profile along the bar major axis with a profile width of 3 pixels.

The extracted radii are converted from pixels to arcseconds and deprojected using the disk PA and inclination. The bar semi-major axis is also deprojected so later analysis can express radius as x/R_bar. A minor-axis profile is extracted only to estimate robust plotting limits from finite positive intensities.

Shoulder detection

Within each galaxy, the full extracted profile is kept for plotting, but the analysis uses only |x/R_bar| < 1.8. The analysed raw intensity profile is interpolated where NaNs occur, normalised to 0..1, and smoothed with a second-order low-pass Butterworth filter. The cutoff is calculated from the number of profile samples and profile span rather than fixed globally.

The algorithm then calculates the first derivative, second derivative, and radius of curvature. Candidate shoulders are found from derivative extrema near each side of the bar: left-side candidates use first-derivative minima and right-side candidates use first-derivative maxima, ordered from the galaxy centre outward. Nearest radius-of-curvature minima and maxima mark the clavicle and shoulder boundaries.

Control Parameters

Parameter	Default	Role
allow_1_sided_shoulders	False	Requires a left and right shoulder pair for a positive shoulder system.
profile_width	3	Pixel width used when extracting major/minor-axis profiles from FITS images.
thin_be_gone	0.05	Rejects shoulders whose inner-to-outer width is too small in x/R_bar units.
peaks_max	12	Rejects profiles with too many first-derivative extrema inside the bar.

Parameter	Default	Role
too_close	0.2	Rejects candidate shoulder clavicles closer than 0.2 R_bar to the centre.
x_cutoff_vs_bre	1.8	Limits the algorithmic analysis window to $\|x/R_{bar}\| < 1.8$.
slope_cutoff	0.35	Candidate shoulder derivative extrema must have absolute slope below this value.
bar_extent	1	Defines the bar boundary in normalised radius after x is divided by R_bar.

Outputs

Output	Description
profiles/<galaxy>_bar-major-axis_profile.dat	Constructed cache of deprojected bar-major-axis radius in arcsec and raw S4G 3.6 micron intensity. Three commented header rows are written.
plots/<galaxy>.png	Diagnostic plot showing the semilog major-axis profile, smoothed curve, derivative overlays, residuals, bar boundary markers, and shoulder annotations.
missing_data_components.txt	Plain-text report explaining which required components are missing or unusable for each skipped galaxy.
missing_data_components.csv	CSV version of the missing-data report with galaxy and reason columns.
shoulder_measurements.npy	Structured NumPy array containing final shoulder measurements.
shoulder_measurements.csv	CSV version of the structured shoulder table.

Output	Description
shoulder_classifications.csv	Per-galaxy SRA classification table with sra_classification, detail text, left/right shoulder flags, failed_extrema, d_extrema, d2_extrema, and roc_minima.
run_summary.txt	Run-level summary of classified galaxies, usable galaxies, skipped galaxies, shoulder systems found, and output locations.
PE_VPD_galaxy_classifications_with_definitions.xlsx	Workbook updated by the merge and formatter scripts. The Classifications sheet keeps isophote PDF links and manual GB visual-review columns near the galaxy name, PE/VPD profile classifications on the left, and SRA output columns on the right.
bar_profile_visual_gallery.html	Local HTML gallery for browsing the isophote profile PDFs with PE/VPD/SRA context, hiding or revealing those classifier labels with the Reveal Classifiers toggle, saving browser-local visual classes and notes, and exporting those visual classifications to CSV.

Measurement columns

Column	Meaning
galaxy	Galaxy name.
clav_left / clav_right	Accepted left/right shoulder clavicle location in x/R_bar units, or NaN if rejected.
left_inner / right_inner	Inner boundary of the accepted shoulder, usually the inner radius-of-curvature minimum around the clavicle.
left_outer / right_outer	Outer boundary of the accepted shoulder, selected from the next radius-of-curvature structure outward.
left_slope / right_slope	First-derivative value at the accepted clavicle candidate.

Column	Meaning
left_clav_inner / left_clav_outer	Radius-of-curvature minima bracketing the left clavicle.
right_clav_inner / right_clav_outer	Radius-of-curvature minima bracketing the right clavicle.

Classification CSV columns

Column	Meaning
galaxy	Galaxy name.
sra_classification	Final SRA label: Shoulders, No Shoulders, Too Noisy, or Missing Data.
sra_classification_detail	Short explanation for the SRA label, such as accepted left and right shoulders, too many extrema, no accepted shoulder pair, or missing-data reason.
left_shoulder_found / right_shoulder_found	Boolean flags showing whether each side survived the candidate filters.
failed_extrema	True when the profile was rejected as too noisy because first-derivative extrema exceeded peaks_max.
d_extrema / d2_extrema / roc_minima	Diagnostic counts used in the plot annotations and classification detail.

Workbook Merge and Formatting Utilities

Companion scripts maintain PE_VPD_galaxy_classifications_with_definitions.xlsx after the SRA run and provide a separate HTML visual-review option. They are separate from the scientific analysis so the workbook and review gallery can be refreshed or restyled without regenerating profiles or plots.

Script	Behavior
Merge Shoulder Classifications Into Workbook.py	Reads shoulder_classifications.csv and updates the Classifications sheet. It also reads the PE and VPD bar-profile classification files, maps raw codes to display labels, writes profile-definition rows to the Definitions sheet, and creates a timestamped backup before saving.
Format PE VPD Classification Workbook.py	Reorders the Classifications sheet so galaxy, isophote PDF links, and manual GB visual-review columns are leftmost, PE/VPD columns follow, and SRA columns are grouped to the right. It applies bold white text on dark-blue headers, freezes the top row, enables filters, auto-sizes columns, styles the Definitions headers, and creates a timestamped backup before saving.
Build Bar Profile Visual Gallery.py	Reads the Classifications sheet and the individual isophote profile PDF folder, then writes bar_profile_visual_gallery.html with one card per galaxy, embedded PDF previews, current PE/VPD/SRA context, browser-local visual class/notes fields, search, CSV export, and a Reveal Classifiers toggle that hides or shows PE, VPD, and SRA together. With --serve it starts a local server that writes visual classes and notes directly back to the workbook.

Visual gallery command-line parameters

Build Bar Profile Visual Gallery.py can be used either to write a static HTML file or to start the workbook-writing local server used by the batch launchers.

Parameter	Default	Role
--pc	Laptop	Selects the Dropbox research-folder layout. Laptop maps to C:\Users\gordo\Dropbox\Public Documents\UCLAN\MSc Research; Desktop maps to D:\Dropbox\Public Documents\UCLAN\MSc Research.

Parameter	Default	Role
--workbook	<research_folder>\Shoulder_Recognition_Erwin\PE_VPD_galaxy_classifications_with_definitions.xlsx	Optional explicit workbook path. Use this only when the workbook is not in the default Laptop/Desktop research-folder location.
--isophote-dir	<research_folder>\Erwin\isophote_output\individual	Optional explicit folder containing the individual <galaxy>_isophote_axes.pdf files shown in the gallery.
--output	<research_folder>\Shoulder_Recognition_Erwin\bar_profile_visual_gallery.html	Output path for the static HTML gallery when --serve is not supplied.
--serve	off	Starts a local HTTP server instead of writing only a static HTML file. In server mode, class and notes changes are saved directly into the workbook.
--host	127.0.0.1	Local network interface used by the server. Keep the default for normal single-PC use.
--port	8765	Port used by the server. The supplied Laptop and Desktop batch launchers override this to 8899, so the browser address is http://127.0.0.1:8899/.

Workbook column group	Columns
Galaxy	galaxy name
Manual visual review	isophote profile PDF, GB visual class, GB visual notes
PE/VPD profile classifications	PE classification, PE profile class, PE profile label, VPD classification, VPD profile class, VPD profile label

Workbook column group	Columns
SRA output	sra_classification, sra_classification_detail, left_shoulder_found, right_shoulder_found, failed_extrema, d_extrema, d2_extrema, roc_minima

Display label	Source profile code(s)
Peak+Sh	BP
Exp	Exp, Exp(N)
Flat-top (FT)	FT, FT(N), Flat-top, Flat-top(N)
Two-slope (2S)	2S, 2S(N), Two-slope, Two-slope(N)

Visual Classification Instructions

The visual-classification workflow is designed to review the individual isophote profile PDFs and write the manual GB classification into the same Classifications sheet that already contains PE, VPD, and SRA outputs. The result is one merged workbook row per galaxy with the manual visual classification beside the classifier outputs.

Before starting

- Close PE_VPD_galaxy_classifications_with_definitions.xlsx before using the gallery server. Excel can lock the file and prevent browser edits from being saved.
- Use the workbook-writing server page, not only the static HTML file, when classifications need to write back to the workbook.
- The Laptop and Desktop launchers start the server on http://127.0.0.1:8899/ and select the corresponding Dropbox research-folder path.

Start the workbook-writing gallery

1. Open PowerShell in C:\Users\gordo\Documents\GitHub\PythonScripts.
2. For the Laptop path, run: & "C:\Users\gordo\Documents\GitHub\PythonScripts\Shoulder Recognition Erwin\Run Bar Profile Visual Gallery Server Laptop.bat"

3. For the Desktop path, run: & "C:\Users\gordo\Documents\GitHub\PythonScripts\Shoulder Recognition Erwin\Run Bar Profile Visual Gallery Server Desktop.bat"
4. Open http://127.0.0.1:8899/ in the browser.
5. Check that the page header shows v2026-06-25 hide-all. If an older page appears, restart the batch file and refresh with Ctrl+F5.

Classify galaxies visually

1. Keep Reveal Classifiers off for a blind visual pass. With the switch off, PE, VPD, and SRA are hidden for every galaxy.
2. Inspect the right-hand major-axis bar-profile plot in the embedded isophote PDF. Use Open PDF if the embedded view is too small.
3. Use the classification picker to select Peak+Sh, Exp, Flat-top (FT), Two-slope (2S), or Unclear.
4. Add any comments in the Notes field.
5. Wait for the page to show Saved to workbook. The selected class is written to GB visual class and the notes are written to GB visual notes in the workbook.
6. Use Export CSV as an optional browser-side backup of the visual classifications.

Merge with classifier results

No separate merge step is needed after using the server page: the workbook is updated in place. Each galaxy row then contains the manual GB visual result, PE/VPD profile classifications, and SRA output columns together.

Result source	Workbook columns
Manual visual review	GB visual class, GB visual notes
PE/VPD classifiers	PE classification, PE profile class, PE profile label, VPD classification, VPD profile class, VPD profile label
SRA classifier	sra_classification, sra_classification_detail, left_shoulder_found, right_shoulder_found, failed_extrema, d_extrema, d2_extrema, roc_minima

- After the visual pass, open PE_VPD_galaxy_classifications_with_definitions.xlsx and use filters to compare GB visual class with PE profile label, VPD profile label, and sra_classification.

- Turn Reveal Classifiers on in the gallery only when checking or comparing against the PE/VPD/SRA outputs.
- If PE/VPD/SRA results are regenerated later, rerun Merge Shoulder Classifications Into Workbook.py and then Format PE VPD Classification Workbook.py. The formatter preserves the GB visual class and GB visual notes columns while restoring the preferred column order and styling.

Detailed Code Documentation

Module setup

The script starts by forcing matplotlib to use the Agg backend, which allows figures to be saved without an interactive display. It then builds PROJECT_ROOT from the script location and appends two sibling folders to sys.path so angle_utils and plot_s4g_isophote_axes can be imported.

- BARPROFILES_DIR: expected to contain angle_utils.py.
- S4G_PLOTTER_DIR: expected to contain plot_s4g_isophote_axes.py.
- The local function required_geometry and profile_at_pa are defined but the active profile-construction path uses corresponding helpers from plot_s4g_isophote_axes.

Utility functions

Function	Behavior
find_nearest(array, value)	Converts the input to a NumPy array and returns the element whose absolute difference from value is smallest. Used to connect derivative extrema to nearest curvature features and profile bins.
read_s4g_table(filename)	Reads whitespace-delimited s4gbars_table.dat with comment lines ignored, applies a fixed 37-column schema, and returns a pandas DataFrame.
read_descramble_map(filename)	Reads scrambled_map.txt, skipping blank/comment lines, and maps integer scrambled IDs to galaxy names.
read_classified_galaxies(filename, descramble_map)	Reads a classification file, keeps rows with a non-'?' classification, resolves the scrambled ID to a galaxy name, and returns the list of classified galaxies.

Function	Behavior
read_manifest(filename)	Reads the geometry manifest CSV into a dictionary keyed by galaxy name.
parse_float(value)	Safely converts manifest fields to finite floats; blank, invalid, and non-finite values become None.
pa_endpoint(pa_deg, radius)	Converts a position angle and radius into an x/y endpoint offset. Defined for profile extraction geometry.
profile_at_pa(data, xc, yc, pa_deg, radius_pix, width)	Extracts a line profile through a 2D image using skimage.measure.profile_line, averaging over the requested linewidth and returning pixel radii plus intensity values.
required_geometry(row)	Parses essential geometry fields from a manifest row, rectifies disk and bar PAs to 0..180 degrees, and returns None if required values are missing.
construct_profile_from_manifest(galaxy, manifest_row)	Builds the profile file, deprojected bar radius, and plot scale for one galaxy, or returns a reason string if the galaxy cannot be processed.
write_missing_data_report(missing_rows, filename)	Writes a human-readable missing-data report that lists required components and per-galaxy failure reasons.

Main sample assembly

After function definitions, the script immediately executes the analysis at module level. It reads the catalog, descramble map, manifest, and both classification files, then constructs classified_galaxies as the sorted unique union of the Peter Erwin and Victor Debattista revised classified samples.

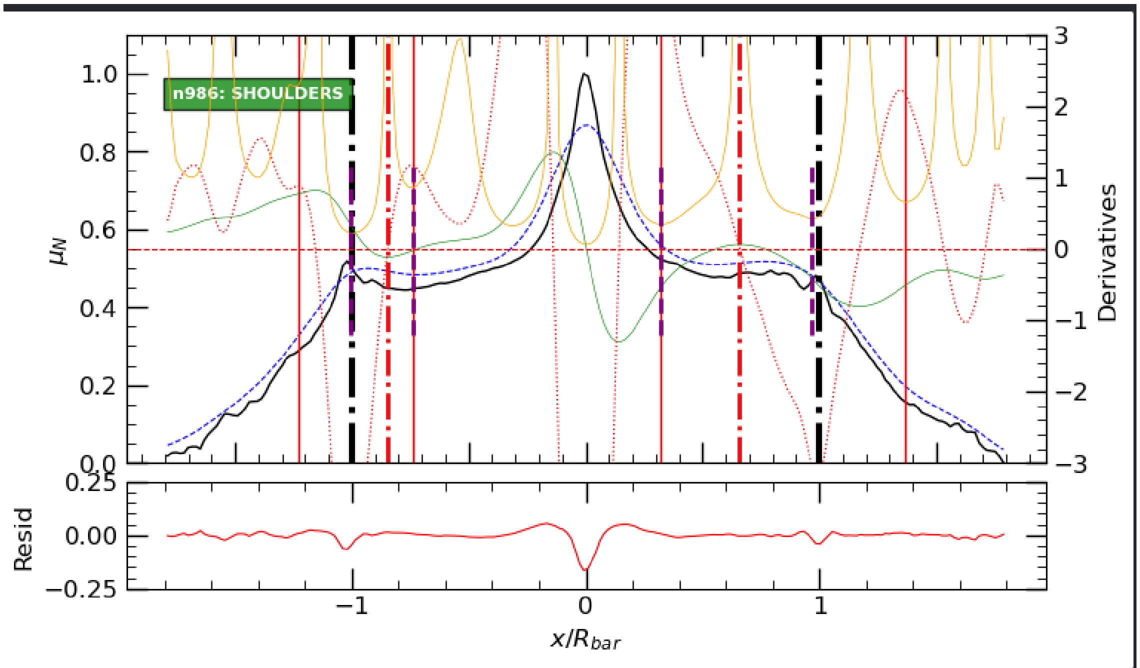
For each galaxy, the code verifies an S4G catalog row, a manifest row, a usable FITS image/profile, and a positive finite deprojected bar radius. Successful galaxies are stored in available_rows with galaxy name, bar_radius, profile_file, and profile_scale; failures are stored in missing_data with an explanatory reason.

Analysis loop details

- Loads each generated profile with np.loadtxt, skipping the three commented header rows.
- Normalises radius by the deprojected bar radius and keeps raw intensity for semilog plotting.
- Repairs leading NaNs by copying the first finite value, then linearly interpolates remaining NaNs with pandas.
- Applies the analysis mask |x/R_bar| < 1.8 and normalises the analysed intensity values to 0..1.
- Computes a Butterworth cutoff from sample count and x-span, then smooths the normalised profile with signal.filtfilt.
- Calculates first derivative, raw first derivative, second derivative, raw second derivative, and radius of curvature.
- Flags a profile as too noisy if the count of first-derivative extrema inside |x| <= 1 exceeds peaks_max.
- Searches left and right candidates separately, starting from candidate extrema closest to x = 0 and moving outward.
- Accepts only shoulders inside the bar, away from the centre, below the slope cutoff, wider than thin_be_gone, and not in a too-noisy profile.
- Rejects final paired shoulders if left and right shoulder regions overlap or are within one bin of the centre.

Plot contents

Each diagnostic PNG has a two-panel layout. The upper panel shows the full raw major-axis intensity profile on a logarithmic y-axis, the smoothed analysed profile over the selected analysis window, derivative overlays on a twin y-axis, bar-boundary markers at x/R_bar = +/-1, and red/purple shoulder markers when accepted. The lower panel shows the residual between the smoothed normalised profile and the normalised raw analysed profile.



Example diagnostic output image included with the script folder.

Operational Notes and Maintenance Risks

- The research-folder location is command-line parameterised with --pc Laptop/Desktop; changing thresholds still requires editing global variables in the source file.
- The code executes at import time. Importing it from another module would run the full analysis unless the main workflow is later moved behind an if __name__ == '__main__' guard.
- Several functions duplicate concepts available in plot_s4g_isophote_axes, and construct_profile_from_manifest currently calls the imported helper versions for geometry/profile work.
- The profile interpolation step includes a note saying it needs discussion, so NaN handling should be treated as an explicit scientific assumption.
- The shoulder decision depends strongly on smoothing, extrema counts, and slope thresholds. These are documented above because they are the most important reproducibility controls.
- The output folder is outside the repository, so generated scientific outputs are not automatically versioned with the code unless copied or tracked separately.

Suggested future improvements

- Extend argparse or add a small configuration file for shoulder thresholds and output options.
- Move the executable workflow into a main() function guarded by if __name__ == '__main__'.

- Write a compact run manifest alongside each result CSV that records parameter values, script version, and input file timestamps.
- Consider replacing print-only progress with logging so rejected candidates can be persisted per galaxy.
- Add unit tests around parse_float, manifest validation, profile construction failure modes, and shoulder candidate rejection logic.